

AACI: The Mechanism for Approximate Authentication and Correction of Images

S. Amir Hossein A. E. Tabatabaei, Obaid Ur-Rehman and Natasa Zivic

Abstract—Modification of images usually originates from channel noise, image processing operations or adversary's efforts. Image authentication mechanisms protect the authenticity of images by differentiating between the effects of malicious and acceptable alterations in the original image.

This paper introduces a mechanism for approximate authentication and correction of images (AACI). The proposed scheme is constructed based on two approximate message authentication code (AMAC) frameworks which use standard cryptographic primitives and error-correcting codes as their building blocks. The AACI tolerates acceptable modifications which are not effective in the image recovery process. The malicious modifications can be detected and the damaged parts can be restored by an AACI mechanism up to a certain limit. In this work, the specification of the AACI mechanism is given, proposed settings, performance and security analysis of the AACI are discussed and simulation results are presented.

Index Terms—Approximate message authentication code (AMAC), image authentication, error-correcting code, unconditional security.

I. INTRODUCTION

Transmission of digital images over non-secure communication networks demands the authentication mechanisms to protect the authenticity of an image data. These multimedia images concern a wide spectrum of applications carrying military, medical, financial, biometric and other information. Image processing operations may introduce some modifications in images which do not influence on their content. There are other factors also which can introduce some non-malicious quantitative modifications of the images data e.g., communication channel noise. Such alterations could be considered as acceptable or permissible modifications depending on the importance of applications. Regarding this issue, the authentication of images could be a difficult task which requires high accuracy and reasonable efficiency. The typical solutions for ordinary data submitted over non-secure networks are given by cryptography and error-correcting codes. Cryptography provides the services for data integrity, confidentiality, authentication and non-repudiation while error-correcting codes enable correcting of erroneous data up to a certain limit and delivering data reliability of decoded data, if soft decoding is used. The essence of acceptable modifications does not exist in the common authentication mechanisms like standard message authentication codes (MAC) which make them disable or at least highly inefficient for image authentication purposes. Also error-correcting codes are not suitable for image authentication

due to their code length issue. There are several techniques of image authentication which are categorized based on the communication model and the type of authentication, each with different approaches [1]–[7]. However, there are no big differences among them and all of them are supposed to satisfy some general requirements like sensitivity, robustness, error localization, portability, restoration and security.

Image authentication approaches consist of two major groups: strict (hard) image authentication and soft image authentication. In strict image authentication no changes in the image content are tolerated. The only accepted manipulations are changes which preserve the pixel value like lossless compression or format conversion [8]. The main techniques used in this group are based on the fragile watermarking the standard cryptography [5], [9]. In soft image authentication the malicious content manipulations are rejected while the rest are accepted [8]. Semi-fragile watermarking and content-based signature are the main approaches in the second group. As a result of increasing number of applications, the research in this field has been led to introducing a generalized concept, approximate message authentication codes (AMAC) [10]. An AMAC refers to the primitive which performs the approximate authentication where the corresponding message set (or tag set) has a fuzzy nature. However, AMACs may not be engaged directly (like standard MACs for binary data) for multimedia authentication purposes without fundamental modifications.

In this paper, a hybrid mechanism for image authentication based on both hard and soft authentication approaches is introduced. The algorithms are based on two generic AMACs but the existing drawbacks in each hinder them to be applied to image authentication solely. The design philosophy of the hybrid mechanism is to protect the most perceptual part of the image data by a hard authentication scheme and the rest by a soft authentication scheme. The hard authentication algorithm is based on the standard MAC mechanism (which does not tolerate any changes in the supported image data) combined by an error-correction code which allows error correction up to a certain limit. Soft authentication algorithm on the other hand, is based on a universal hash function [11] and accepts some pixel modifications on high-frequency components of the image data which do not influence the image recovery operations. The error correction capability, the accuracy and the security level of the hybrid mechanism are adjusted depending on the image sensitivity and its application. The introduced approach is called an approximate authentication and correction of images (AACI) mechanism.

Section II gives briefly the required definitions and background. The principal AMAC algorithms along with necessary mathematical tools in image processing are recalled as well.

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The algorithms specifications are discussed in Section III. Section IV contains the security and performance analysis. The implementation and simulation results are given to validate the theoretical proofs. Section V concludes the paper with suggestions for generalizing and improving of the proposed mechanism.

II. APPROXIMATE MESSAGE AUTHENTICATION CODES AND IMAGE AUTHENTICATION

A. Definitions

In a standard MAC scenario, any changes of the input message data are reflected by around 50% complemented bits in its authentication tag. In contrast to standard MACs, AMACs are proper primitives where a modest number of changed bits in the received message is tolerated. The concept of soft authentication which is followed in the latter scheme indicates a level of authenticity i.e., approximate authentication. In approximate message authentication schemes the acceptable changes in a message are reflected by recognizable modifications in its authentication tag [10], [12]. The distance function d (here Hamming distance) together with a threshold value δ is used to introduce approximate authentication by verifying the correctness of the received message having up to (acceptable) δ differences from the original message. This property imports new frameworks for security and accuracy analysis in the formal definition. In the following definition which is the formalized dressed up definition of “approximately secure and approximately correct message authentication code for distance function d ” in [10] and “AMAC” in [13], $\mathcal{K}$ is the key set, $\mathcal{M}$ is the message set and $\mathcal{T}$ is the tag set.

Definition 1. A (d, p, δ) -approximate correct and $(d, \gamma, t, q, \epsilon)$ -approximate secure message authentication code is a tuple $(\mathcal{K}, \mathcal{M}, \mathcal{T}, d, \text{Kg}, \text{Tag}, \text{VF})$ such that the following requirements are satisfied:

- 1) d, Kg, Tag and VF are four polynomial-time algorithms.
- 2) $\text{Kg}(\cdot, d) : \{0, 1\}^l \rightarrow \{0, 1\}^l$ is a key generator algorithm which generates an l -bit key on l -bit input.
- 3) $\text{Tag} : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ is the tag generator algorithm which generates an authentication tag on input message and key.
- 4) $\text{VF} : \mathcal{K} \times \mathcal{M} \times \mathcal{T} \rightarrow \{0, 1\}$ is a verification algorithm such that:

$$\forall k \in \mathcal{K}, \forall m \in \mathcal{M}, \text{Tag}(k, m) = t \iff \text{VF}(k, m, t) = 1.$$

- 5) (d, p, δ) -Approximate correctness: $\forall m, m' \in \mathcal{M}, \forall k \in \mathcal{K}, \text{Prob}\{\text{VF}(k, m', t) = 1 \mid d(m, m') \leq \delta, \text{Tag}(k, m) = t\} \geq p$.
- 6) $(d, \gamma, t, q, \epsilon)$ -Approximate security: $\forall (m_i, t_i) \in \mathcal{M} \times \mathcal{T}, i = 1 \dots q : \text{Tag}(k, m_i) = t_i, \text{Prob}\{\text{VF}(k, m, t) = 1 \mid d(m, m_i) \geq \gamma\} \leq \epsilon$.

To consider the security strength of the mechanism two additional properties are introduced depending on the adversary model: weak preimage resistance and strong preimage resistance [10]. Skipping formal definitions, the weak preimage resistance states that an adversary with sufficient number of valid pairs $(m_i, \text{Tag}(k, m_i))$ has a negligible chance

to find message m' which passes the verification test i.e., $\text{VF}(k, m', t_i) = 1$. If the additional assumption of accessibility of adversary to the secret key k is given, the latter property is called strong preimage resistance.

B. Two Approaches to Approximate Secure and Approximate Correct MAC

Constructional models for AMAC have been introduced in [10], [13] and [14]. Among them, two approaches which are used in the proposed image authentication scheme will be briefly recalled. In the weak preimage resistant scheme [10], the authentication tag is created based on a symmetric encryption algorithm (E_k, D_k) , a systematic error-correcting code (Enc, Dec) and a standard MAC algorithm $\text{MAC}(k, \cdot)$. E_k (D_k) indicates encryption (decryption) algorithm and Enc (Dec) indicates encoding (decoding) algorithm respectively. The error correction capability of (Enc, Dec) is δ . This scheme is a (d, p, δ) -approximate correct and $(d, \gamma, t, q, \epsilon)$ -approximate secure message authentication code with optimal parameters $p = 1$ and $\gamma = \delta + 1$ [10]. The tag generation algorithm is given as follows.

Algorithm 1 AMAC₁ tag generation [10]

Input: Shared secret keys k_1, k_2 generated by $\text{Kg}(\cdot, d), m$ -bit Message x , Error correction capability δ

- 1: **procedure** AMAC₁ – Tag(k, x)
 - 2: $x||pc = \text{Enc}(x)$ $\triangleright pc$ indicates parity check bits.
 - 3: $t_1 \leftarrow \text{MAC}(k_1, x)$
 - 4: $C \leftarrow E_{k_2}(pc)$
 - 5: $tag \leftarrow C||t_1$
 - 6: **return** tag
 - 7: **end procedure**
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The verification procedure part of AMAC₁ is standard (see [10]). In the second algorithm, a strong preimage resistant scheme, the authentication tag is generated based on the construction of probabilistic universal one-way hash functions [11]. In the description of the tag generation algorithm, $\text{PRNG}_k(\cdot)$ and π indicate a pseudo-random number generator and a pseudo-random number permutation respectively. Also the finite universal one-way hash function is denoted by $\text{UOWHF}_u(\cdot)$.

The output tag length of Algorithm 2 is t_2b bits where $|t_{hash_i}| = b$ bits. The corresponding verification algorithm differs from standard MACs due to approximate correctness property. In this scheme, the verification algorithm checks for equality between the received and the computed hash-block sequences in a defined number of positions. The verification procedure will be presented in Section III as a part of whole verification algorithm. The latter scheme addresses the necessities of AMACs which may be commonly appeared in fuzzy-nature communication applications. However in some examples like image authentication, additional requirements (II-C, III and [5]) may be demanded which are not satisfied by each scheme individually.

Algorithm 2 AMAC₂ tag generation [10]

Input: Shared secret key k generated by $Kg(., d)$, Initial value IV , m -bit message x , Block size c , Number of blocks t , Approximate correctness parameters p and δ , Approximate security parameter γ

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1:  $t_1 = \lceil m/2c\delta \rceil, t_2 = \lceil -10 \log(1-p) \rceil$ 
2: procedure AMAC2 – Tag( $k, x$ )
3:    $(u || \pi || \rho || L) \leftarrow PRNG_k(IV)$ 
4:    $u \in \{0, 1\}^k, \pi \in \{0, 1\}^{\lceil \log_2(t_1) \rceil}, L \in \{0, 1\}^m$ 
5:    $x_1 || \dots || x_t \leftarrow \pi(L \oplus x)$ 
6:   for  $i = 1, t_2$  do
7:      $N_i \leftarrow x_{i_1} || \dots || x_{i_{t_1}}$ 
8:      $hash_i \leftarrow UOWHF_u(N_i)$ 
9:   end for
10:   $tag \leftarrow IV || hash_1 || \dots || hash_{t_2}$ 
11: return tag
12: end procedure

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C. Image Authentication

The typical procedure in image authentication schemes works as follows. To authenticate an image, the authentication tag is calculated by the sender based on the shared key with the receiver and appended to/in the compressed image. The receiver decompresses the image which results in an approximate image supposed to be very close to the original one [13]. The fuzziness nature of the image space and noise tolerance property of the algorithm, are combined to result in a successful verification.

Involving image processing operations (e.g., compression, geometrical transformations, scaling, etc.) in image authentication methods requires mathematical tools like orthogonal transforms and subband coding [15]. In image compression techniques, at first the correlation between pixels is reduced and then quantization and entropy coding are performed. Orthogonal transform like discrete cosine transforms (DCT), Karhunen-Loeve transform (KLT) and subband coding like discrete wavelet transform (DWT) are the most well-known methods applied in decorrelation step. The DCT is used in the given approach in this paper. To apply the DCT to an image, after dividing the image into square blocks, 2-D forward DCT is performed which results in a square matrix. In derived matrix the upper most left element which corresponds to zero frequency is the DC coefficient while the rest are the AC coefficients. Due to the sensitivity of a human visual system, in the DCT-based image authentication schemes the DC coefficient and important ACs are supported by DCT-based authentication algorithms for verification and restoration of damaged parts [5]–[7].

III. THE HYBRID APPROACH FOR APPROXIMATE AUTHENTICATION OF IMAGES

The proven security and correctness properties of Algorithms 1 and 2 are shown in [10]. In the first algorithm the tag length is influenced by the performance of the systematic error-correcting code which is not desirable in the practical applications where limitation on the tag length is imposed.

The second algorithm allows approximate correctness in the received message but not restoration capability. In this section, a hybrid scheme for robust authentication of images is given which enables restoration and partial error localization. The proposed mechanism uses Algorithms 1 and 2 as its components followed by some adjustments.

A. Approximate Authentication and Correction of Images (AACI)

The algorithm for calculating the authentication tag in an AACI scheme is identical for both sender and receiver: The image I which is supposed to be transmitted over a non-secure channel is divided into $M \times N$ disjoint blocks each of $m \times m$ pixels size. $m = 8$ is a typical option as it is compatible with the JPEG compression where 8×8 blocks are subjected to the 2-D forward DCT [1], [15]. The DCT is applied to each pixel block and the DC element of each matrix is picked, which results in an lMN -bit sequence where l is the length of each represented element of the DCT matrix. This sequence contains the substantial energy of the image and is generally more affected by content manipulation of the image than high frequency components [16]. Therefore, this sequence demands the maximum protection and no error tolerance. The first algorithm AMAC₁ is used for the tag generation of the DCs bit-sequence. The corresponding verification algorithm enables error detection and possible correction. The tag generated by AMAC₁ is denoted by st_1 .

To generate a final tag, the first ten AC coefficients of the DCT matrix of each block are selected according to zigzag order used in the JPEG compression. The resulted $10l$ -bit sequence is considered as a message block (out of MN blocks) according to Algorithm 2. The second algorithm AMAC₂ is used to generate the authenticate tag of the $10lMN$ -bit message which results in subtag st_2 . The tunable parameters p, δ and γ are chosen according to desired accuracy and security level. Protecting the first ten AC coefficients along with the DC coefficient which have sufficient information for block restoration ensures acceptable security against common types of image tampering [8]. The final tag is a concatenation of two subtags i.e., $tag = st_1 || st_2$. At this point the sender transmits the compressed image I over noisy channel and its generated tag, tag , via reliable channel which results in a received noise-free tag. The formal description of the tag generating algorithm is shown in Algorithm 3.

The verification process of the AACI is as follows: the receiver gets the compressed image with the attached AACI tag. The image is decompressed and the AACI tag is computed on the image based on the shared secret keys and compared to the received one. If the verification succeeds on the first subtag st_1 which means the successful hard verification on the MAC part, then the verification on the second part st_2 starts. If the verification on the first part is unsuccessful then the detected errors are tried to be corrected by the error-correcting code. The errors can be corrected when they are not beyond the capability of error-correcting code. Otherwise the verification algorithm stops and authentication fails. The verification procedure on the second subtag is soft and it

Algorithm 3 AACI tag generation

Input: Image I , Image dimensions M, N , Error correction capability of $AMAC_1$ δ_1 , Shared secret keys k_1, k_2 , Initial value IV , Approximate correctness parameters p and δ_2 , Approximate security parameter γ

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1: Let  $\{b_{1,1}, \dots, b_{M,N}\}$  be image blocks of  $I$ .
2: procedure AACI – Tag( $k_1, k_2, I$ )
3:   for  $i = 1, M$  do
4:     for  $j = 1, N$  do
5:        $A = \text{DCT}(b_{i,j})$   $\triangleright$  DCT applies to each block.
6:        $DC_{i,j} \leftarrow A_{1,1}$   $\triangleright$  DC of each block is
         extracted.
7:        $AC_{i,j} \leftarrow A_{1,2} || A_{2,1} || \dots || A_{5,1}$   $\triangleright$  ACs of each
         block are extracted.
8:     end for
9:   end for
10:   $M_1 = DC_{1,1} || DC_{1,2} || \dots || DC_{M,N}$ 
11:  Let  $M_2 = \{AC_{1,1}, \dots, AC_{M,N}\}$ 
12:   $st_1 = \text{AMAC}_1 - \text{Tag}(k_1, M_1)$ 
13:   $st_2 = \text{AMAC}_2 - \text{Tag}(k_2, M_2)$ 
14:   $tag = st_1 || st_2$ 
15:  return  $tag$ 
16: end procedure

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depends on the tunable parameters which indicate the level of accuracy and security. The verification in the second phase will be successful if at least $\lceil Tr.t_2 \rceil$ number of positions in st_2 have equal hash blocks. The choice for threshold ratio Tr will be discussed in Section IV-A.

The formal description of the AACI verification algorithm is given in Algorithm 4. The output of the verification algorithm of the AACI indicates an approximate authentication of the images where the level of approximate accuracy and security are introduced by δ, δ_2 and p which are initialized depending on the application. The main existing obstacle for implementation of the AACI is the tag length of algorithm $AMAC_1$ which depends on the bit-sequence length. In the following section, practical solutions are suggested for this problem.

B. The Choice of Error Correcting Code

Reed Solomon (RS) codes [17] are non binary cyclic error-correcting codes. They are defined over a Galois Field ($GF(p^m)$) where p is a small prime number and is typically chosen as 2 while m is a small integer and typically $2 \leq m \leq 8$. RS codes have the ability to detect and correct multiple symbol errors. They are extremely popular due to their excellent burst error correction capability. Aside from correcting a burst of errors, they can also correct random symbol errors. If t -symbol-error correction capability is desired, then $2t$ parity symbols are calculated and appended to the data symbols to make an RS codeword of the $RS(n, k, d_{min})$ code. Here n is the codeword length, k is the data length and d_{min} is the minimum Hamming distance of the code. Thus up to $2t$ symbol errors can be detected but up to t symbol errors can be corrected, where each symbol has p^m bits. For the chosen application, i.e., images, RS codes are a perfect choice. As

Algorithm 4 AACI verification

Input: Decompressed received image I' , Image dimensions M, N , Received tag , Shared secret keys $k_1 = k_{1,a} || k_{1,e}, k_2$, Initial value IV, δ_1 , Approximate correctness parameters p and δ_2 , Threshold ratio Tr , Approximate security parameter γ

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1:  $t_1 = \lceil MN/2\delta_2 \rceil, t_2 = \lceil -10 \log(1-p) \rceil$ 
2: Let  $\{b'_{1,1}, \dots, b'_{M,N}\}$  be image blocks of  $I'$ .
3: procedure AACI – VF( $k_1, k_2, I'$ )
4:   for  $i = 1, M$  do
5:     for  $j = 1, N$  do
6:        $A = \text{DCT}(b'_{i,j})$ 
7:        $DC'_{i,j} \leftarrow A_{1,1}$ 
8:        $AC'_{i,j} \leftarrow A_{1,2} || A_{2,1} || \dots || A_{5,1}$ 
9:     end for
10:  end for
11:   $M'_1 = DC'_{1,1} || DC'_{1,2} || \dots || DC'_{M,N}$ 
12:  Let  $M'_2 = \{AC'_{1,1}, \dots, AC'_{M,N}\}$ 
13:  Write  $tag = st_1 || st_2$ .
14:  Let  $M''$  be the extracted message from  $st_1$ .
15:  if  $MAC(k_{1,a}, M'_1) = MAC(k_{1,a}, M''_1)$  then
16:    go to 24
17:  else
18:    if Error correction on  $M'_1$  is possible then
19:      Perform error correction and go to 15
20:    else
21:      Halt and return 0
22:    end if
23:  end if
24:   $st'_2 = \text{AMAC}_2 - \text{Tag}(k_2, I')$ 
25:  Write  $st'_2 = IV || thash'_1 || \dots || thash'_{t_2}$  and  $st_2 =$ 
     $IV || thash_1 || \dots || thash_{t_2}$ 
26:  if  $thash_i = thash'_i$  for at least  $Tr.t_2$  number of blocks
    then
27:    return 1
28:  end if
29:  return 0
30: end procedure

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explained in Section II-C, the image is divided into rows and columns of blocks and the DCT is calculated for each block. The DC elements are used by $AMAC_1$. Let N be the number of blocks in a row and M be the number of blocks in a column, then there are MN DC elements. Each row of the DCs can be protected using RS codes. Thus M codewords are generated, each of n symbols. Each DC element needs to be represented as an RS symbol. For a grey-scale image, the DC elements can be represented in 8-bits ($l=8$), and therefore $RS(255, 223, 33)$ code is chosen for encoding in $AMAC_1$. The chosen RS code has the ability to correct 16 symbol-errors. For a 128×128 -pixel image, N is 16 but the required k for $RS(255, 223, 33)$ is 223. Thus code shortening has to be used, i.e., 207 “0” symbols are appended to the N DC symbols to complete the data block and then the RS encoding is performed to obtain a codeword. The process is repeated for all the rows of the DC elements resulting in the M RS codewords. $AMAC_1$ uses only the parity symbols in encryption. Optional interleaving

can be performed on the rows and the columns of the DC elements for additional robustness against transmission errors. A simple example would be to write the rows of DC elements as bits and transmit the data column after column. This would spread the error burst to multiple codewords.

IV. ANALYSIS AND SIMULATION OF AACI

A. Security Analysis

Security analysis of the AACI is based on the proven facts of principal components in [10]. The AACI scheme protects the most perceptual parts of the compressed image by an algorithm which provides approximate security and approximate correctness. This part is weak preimage resistant. However, error correction is possible only in this phase due to the importance of the protected part. The less important part of the image is supported by the approximate secure and approximate correct MAC which provides strong preimage resistance property. To consider the performance of the AACI approach, the following random variables are defined:

$$NB_i = \begin{cases} 1 & \text{if } \text{hash}_i \neq \text{hash}'_i \\ 0 & \text{if } \text{hash}_i = \text{hash}'_i \end{cases}, \quad NB = \sum_{i=1}^{t_2} NB_i$$

Using classical occupancy problem [18], the probability of appearing one non-matched block, P_{NB} , in the second subtag is calculated as follows (remind that $|\text{hash}_i| = b$ and there are MN image blocks).

$$\begin{aligned} P_{NB} &= P(NB_i = 1) \\ &= 1 - \left(1 - \frac{\binom{MN - t_1 + \delta_2 - 1}{\delta_2}}{\binom{MN + \delta_2 - 1}{\delta_2}}\right) 2^{-b} + \frac{\binom{MN - t_1 + \delta_2 - 1}{\delta_2}}{\binom{MN + \delta_2 - 1}{\delta_2}} \end{aligned} \quad (1)$$

The security analysis here is proceeded in the unconditionally secure framework. In this viewpoint, the potential threats are impersonation and substitution attack. In an impersonation attack the attacker tries to deceive the receiver with the spoofed message and a tag pair (I, tag) of his choice while in a substitution attack an accessible authentic message is substituted by the of adversary's choice. To mount an attack successfully on the AACI, the attacker has to pass two-phase verification with a spoofed message (here, an image). The first phase, $AMAC_1$, tolerates no errors which imposes an attack against standard MAC on the attacker. However, the second part, $AMAC_2$, may tolerate few errors. Assuming P_I and P_S as the successful deception probabilities in an impersonation attack and a substitution attack respectively, their lower bounds are obtained by expanding the results of [13]. Two adversary models are considered in the analysis of substitution attack: active model and passive model. In an active adversary model, the adversary can interact actively with the parties for tampering the communication. In this model the adversary has access to the tag generation algorithm as a block-box oracle and can get the tag of a queried message. The lower bound is common for P_I and P_S in this model:

$$\{P_I, P_S\} \geq p \frac{\sum_{i=0}^{\lceil 0.5 P_{NB} t_2 b \rceil} \binom{t_2 b}{i}}{2^{t_2 b}} \quad (2)$$

In a passive adversary model, the adversary can just eavesdrop or sniff over communication and has access to an authenticated message-tag pair submitted by the sender over the channel. The probability distribution of the messages, $P_{\mathcal{X}}$, should be taken into account in a passive adversary model as the adversary can not influence the distribution of message-tag pairs (it is independent of adversary's effort). The lower bound for deception probability of a substitution attack in an active adversary model where an authenticated message x is supposed to be spoofed by the adversary is computed as follows.

$$P_S \geq p P_{\mathcal{X}}(x) \left(2^{t_2 b} - \frac{\sum_{i=0}^{\lceil 0.5 P_{NB} t_2 b \rceil} \binom{t_2 b}{i}}{2^{t_2 b}}\right) \quad (3)$$

More smooth lower bound is obtained by summation over all captured authentic message-tag pairs (denoted by $\mathcal{X}^*$) over the communication channel by the adversary:

$$P_S \geq p \left(2^{t_2 b} - \frac{\sum_{i=0}^{\lceil 0.5 P_{NB} t_2 b \rceil} \binom{t_2 b}{i}}{2^{t_2 b}}\right) \sum_{x \in \mathcal{X}^*} P_{\mathcal{X}}(x) \quad (4)$$

In this viewpoint the overall deception probability is obtained by finding a maximum between two probabilities i.e., $P_D = \max\{P_I, P_S\}$. To obtain a desirable level of security and accuracy, it is possible to find a lower bound for the number of required hash-blocks by initializing the other adjustable parameters. Therefore, the probability of appearing less number of non-matched blocks than a threshold value ($\lceil Tr.t_2 \rceil$) must be calculated. The corresponding random variable NB follows the binomial distribution with parameters t_2 and P_{NB} . So after applying Chernoff inequality [10], [19] ($Pr(NB < Tr.t_2) \leq e^{-2t_2(P_{NB} - Tr)^2}$) and approximate accuracy requirement ($Pr(NB < Tr.t_2) \leq 1 - p$), a lower bound for number of required hash blocks [10] or the threshold ratio can be extracted smoothly:

$$Tr \geq P_{NB} - \sqrt{\frac{1}{2t_2} \ln \frac{1}{1-p}} \quad (5)$$

A potential threat in the hybrid scheme AACI is the impact of image tampering. The verification could be successful on a valid tampered image which is not authentic if the tampered pixels are not reflected by the protected DCT components. Protecting eleven DCT matrix elements of each image block makes this attack scenario nearly impossible. However, it is not proven for the AACI and comprehensive analysis regarding to the impact of image tampering is required.

B. Implementation and Simulation Results

128×128 pixel gray-scale images are chosen in the simulations. 8×8 pixel blocks are selected, giving $M = N = 16$. This gives 16×16 DC elements, and therefore 16 RS codewords are generated. Since data is 16 symbols long, $RS(32, 16)$ codeword is obtained by shortening an $RS(255, 223)$ codeword. Thus all the symbol errors in the data block (or the DC elements) can be corrected for the chosen image resolution. For higher resolutions, some of the DC elements might not be correctable if the errors exceed 16 symbols.

The HMAC-SHA256 and the AES-256 are selected for implementing the cryptographic components as the corresponding MAC and the symmetric encryption algorithms respectively. The tag length can be shortened by truncating of the generated hash-blocks up to a limit where the security requirements are not threatened. The simulations are performed over additive white Gaussian noise (AWGN) channel with binary phase-shift keying (BPSK) modulation and Turbo Codes of rate 1/3. The results are shown at E_b/N_0 of 2.5dB. The source image is shown in Figure 1 followed by the received image in Figure 2. The possible error blocks are identified and shown in Figure 3. The same (potential) error locations in the received image are shown in Figure 4. Finally the corrected image is shown in Figure 5. The comparison between image error rates obtained in both standard authentication and the AACI authentication is shown in Figure 6.



Fig. 1: Source image

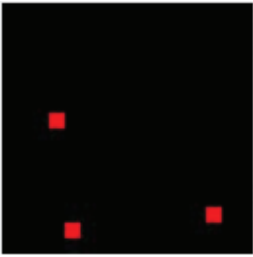


Fig. 3: Located erroneous blocks



Fig. 5: Corrected image



Fig. 2: Received image



Fig. 4: Erroneous blocks shown in the image

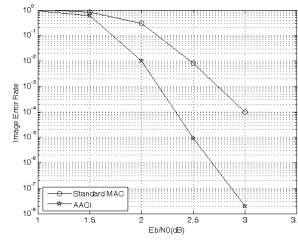


Fig. 6: Image error rate

V. CONCLUDING REMARKS

An algorithm for approximate authentication and correction of images based on the standard MAC and the error-correcting code was given. The security analysis has been presented in the unconditionally secure viewpoint and the simulation

results on a gray-scale image have been shown. More possible improvement includes:

- Generalizing the AACI mechanism to the DWT domain: this enables the applicability to the JPEG2000 compression technique.
- Improving the error localization and error detection capabilities: this can be done by efficiently modifying the row-column based or the block-based methods for example.
- Involving the statistical distribution of frequency components in the analysis: it results to obtaining more smooth lower bounds in security analysis which can increase the performance and security of the AACI.

The above points will be considered on the given approach for further improvements.

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